

Operating Conditions and Performance Standards

Stamping press 01 (press-stamp-01) | Main drive and lubrication circuit

Machine	Stamping press 01 (machine ID press-stamp-01, type press, area Press shop, site vit-lab), stamping press in a car plant
Sensors	MAIN_MOTOR_CURRENT (A), LUBE_OIL_PRESSURE (bar), BEARING_VIBRATION_RMS (mm/s) and the 64-bin spectrum BEARING_SPECTRUM (mm/s); sensor IDs PRESS-STAMP-01.<CHANNEL_KEY>
Components / subsystem	Press main drive and its pressure lubrication circuit: main drive motor, lubrication unit (gear pump, reservoir, lube filter) and main bearing housing; planned agent tags main_drive_motor, lube_filter, lubrication_unit, main_bearing_housing
Document title	Operating Conditions and Performance Standards
Document ID	PS01-DOC-01
Revision / date	1.0 / 2026-10-08
Purpose	Defines normal operation of press-stamp-01 (main drive motor current, lube oil pressure, bearing vibration RMS and the bearing spectrum) and the limits needed to decide whether telemetry is normal, watch, warning, alarm, critical or a sensor and data problem.
Scope	Normal behaviour, ranges, limits, operating states, load dependence, environment, transients, rate of change and alert escalation for the three scalar channels and the spectrum. Fault signatures are in Document 02, the diagnostic procedure in Document 03, maintenance in Document 04 and simulated cases in Document 05.

Provenance tags used throughout this document

[P]	Project value: taken from the repository or CONTEXT_PRESS_STAMP_01.md (catalog limits, simulation equations, topics, runbook stubs, tests).
[D]	Derived value: computed from the repository's own simulation model (seed 1, 2 Hz, 600 s per case) or algebraically from its equations, as recorded in the context file.
[E]	Engineering reference: general industrial practice for press drives, lubrication circuits and vibration diagnostics. Not a manufacturer value and not taken from the repository.
[I]	Illustrative / proposed prototype value: chosen by the authors for this knowledge base and meant to be refined with real plant data.

Important: the telemetry for this machine is simulated. There is no physical press or transducer. An ESP32 gateway (or a software simulator) plays the press controller and generates values from documented equations. The repository describes its limits as illustrative constants chosen for the demo, not vendor values, and its runbooks as not an OEM manual. Values tagged [P] are therefore prototype values, not OEM specifications; values tagged [E] and [I] are not manufacturer-certified.

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1. Document Control and Retrieval Guide

1.1 Document set

Table 1. The five-document set for press-stamp-01

Doc	ID	Title	Question it answers	Use it when
01	PS01-DOC-01	Operating Conditions and Performance Standards	What is normal, and where do warning, alarm and critical begin?	A reading must be classified
02	PS01-DOC-02	Failure Modes and Fault Signatures	What does each fault look like in the telemetry?	A candidate fault list is needed
03	PS01-DOC-03	Diagnostic Procedures and Decision Logic	How do I go from the observation to a diagnosis and action?	A decision must be justified
04	PS01-DOC-04	Maintenance, Inspection and Troubleshooting	What do I inspect, repair and verify?	A work order is prepared
05	PS01-DOC-05	Fault History, Case Scenarios and Diagnostic Evidence	How did similar situations resolve?	A pattern must be compared with worked examples

1.2 Key identifiers used for retrieval

Table 2. Identifiers

Item	Value	Tag
Machine ID	press-stamp-01 (label Stamping press 01, machine type press, typeCode 2, area Press shop, site vit-lab)	[P]
Sensor IDs	PRESS-STAMP-01.MAIN_MOTOR_CURRENT, PRESS-STAMP-01.LUBE_OIL_PRESSURE, PRESS-STAMP-01.BEARING_VIBRATION_RMS	[P]
Spectrum key	BEARING_SPECTRUM, 64 bins, startHz 0, stepHz 12.5, intervalMs 2000, fundamentalHz 24.7	[P]
Stroke cycle	4.6 s per stroke (about 13 strokes per minute); working stroke 38 to 60 percent of the stroke	[P]
Publish interval	Telemetry 500 ms (adjustable 250 to 5000 ms); spectrum 2000 ms	[P]
Fault scenarios	NORMAL, CLOGGED_FILTER, BEARING_WEAR (ramp r from 0 to 1 over rampSec, default 120 s, allowed 5 to 600 s)	[P]
Topic root	cdo/v1/vit-lab/press-stamp-01/{birth, telemetry, spectrum, status, diag, log, cmd/*, ack}	[P]
Planned agent component tags	main_drive_motor, lube_filter, lubrication_unit, main_bearing_housing	[P] planned
Suggested twin meshes	090-stamping-press/0 (current), 090-stamping-press/1 (pressure), 090-stamping-press/ram/steel (vibration); suggestions only, no binding captured	[P]

1.3 How to use this document

- For a single reading, go to Table 17 (master limits) and Table 18 (open and clear rules). A reading is classified on a 10 s window, not on a single sample (Section 6.3).
- For a trend, use the rate-of-change and time-to-threshold tables in Section 16.
- For a pattern across channels, use Document 02 (signatures) and Document 03 (decision logic). This document only classifies levels.
- Every number carries a provenance tag. A [P] number is a prototype value from the repository, not an OEM specification.

RULE: never describe a [P] limit as a manufacturer value. The repository calls its limits "illustrative constants chosen for the demo, not vendor values".

1.4 Terms and abbreviations

Table 3. Terms

Term	Meaning
I, P, V	Motor current (A), lube oil pressure (bar), bearing vibration RMS (mm/s)
r	Fault ramp, 0 to 1, linear over rampSec, then held. Published only in diag.sim (ground truth); diagnostic reasoning uses channel evidence instead
V60, P60, I60	60 s window means of V, P and I (120 samples at 2 Hz)
V10, P10, I10	10 s window means (20 samples)
Ipk	Peak current in one 4.6 s stroke
PF60	Fraction of samples in the last 60 s that are at or beyond the warning limit
HB300	Mean spectrum bin amplitude above 300 Hz (bins 24 to 63)
PMR300	Peak-to-median ratio of the bins above 300 Hz
1X, 2X	Shaft frequency 24.7 Hz and its second harmonic (about 49 Hz)
Levels	L0 normal, L1 watch, L2 warning, L3 alarm, L4 critical, LS sensor or data suspect

2. Machine and Subsystem Overview

2.1 Machine

press-stamp-01 is a stamping press in a car plant (press shop, site vit-lab) [P]. The monitored subsystem is the press main drive and its pressure lubrication circuit [P]. The repository does not state the press make, model, tonnage, drive type, die or serial number [P]; none is assumed here.

Table 4. Machine data

Field	Value	Tag
Machine type	press (typeCode 2)	[P]
Stroke time	4.6 s (sim.cycleSec); about 13 strokes per minute	[P]
Working stroke	p = 0.38 to 0.60 of the stroke, that is 1.75 to 2.76 s into each 4.6 s cycle; load is a sin-squared pulse over this window	[P] / [D] seconds
Main motor speed	About 1480 rpm; shaft frequency 24.7 Hz (1X)	[P]
Gateway	ESP32 running cdo-edge-gateway with MACHINE_TYPE 2, firmware 1.0.0 in contract examples; stands in for a real controller reached by OPC UA or MTConnect	[P]
Make, model, tonnage, die, oil grade, bearing type	Not specified in repository	[P]

2.2 Lubrication circuit and sensor placement

A gear pump draws oil from the reservoir and pushes it through the lube filter to the main bearings. The pressure transducer is fitted downstream of the filter, in the distribution line that feeds the bearings [P]. Consequence: a blocked filter makes the pressure reading fall. A rising reading would point to a blockage in a nozzle or line after the transducer, which is a different fault and is not modelled [P].

Table 5. Flow path and what each sensor sees

Element	Role	Sensor effect
Reservoir	Oil supply	Level not measured [P]
Gear pump	Produces flow; causes 1 Hz pulsation of about 0.12 bar [P]	Visible as pressure ripple
Lube filter	Removes particles	Clog lowers downstream pressure (P falls up to 45 percent)
Pressure transducer	Reads distribution line downstream of the filter	LUBE_OIL_PRESSURE
Main bearings	Carry the main shaft; need an oil film	Starved film raises vibration (BEARING_VIBRATION_RMS rises)
Main drive motor	Drives the press at about 1480 rpm	MAIN_MOTOR_CURRENT, friction raises it slightly

2.3 Sensor-to-component relationships

Table 6. Channels, components and twin binding

Channel	Component (binding label)	Planned component tag	Suggested mesh	Binding sensorType
MAIN_MOTOR_CURRENT	Main drive motor	main_drive_motor	090-stamping-press/0	current
LUBE_OIL_PRESSURE	Lubrication unit	lube_filter, lubrication_unit	090-stamping-press/1	pressure
BEARING_VIBRATION_RMS	Main bearing housing	main_bearing_housing	090-stamping-press/ram/steel	vibration
BEARING_SPECTRUM	Same vibration seen by frequency	none	no binding (not bindable)	n/a

The twin asset (Car Factory, Assembly Hall, car_factory.glb) contains a stamping press but no lubrication unit, drive motor or bearing housing as separate parts. An operator binds a mesh and the label says what it stands for. docs/demo-bindings.json records meshName null for all three channels, so no binding has been captured [P]. One mesh carries one sensor.

Causal link stated in the repository: a clogged filter starves the bearings, so ignoring it is how bearing damage eventually happens [P]. The channels are therefore related: LUBE_OIL_PRESSURE can lead BEARING_VIBRATION_RMS in a real degradation, although in the simulator the clogged-filter vibration rise is imposed directly.

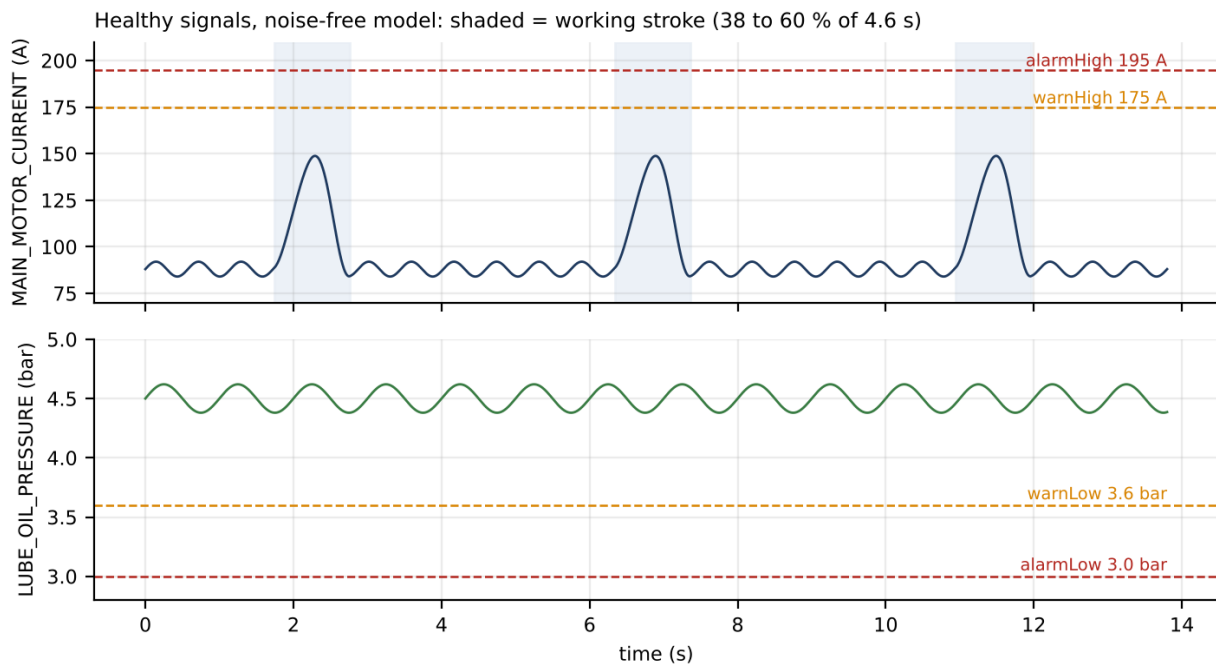


Figure 1. Healthy MAIN_MOTOR_CURRENT and LUBE_OIL_PRESSURE from the noise-free equations [D]. The current pulse peaks near 150 A in the working stroke; the pressure ripple is 0.12 bar at 1 Hz.

3. Operating States

The repository models one running machine state. Other states exist in a real press and are listed as engineering reference [E] so the agent can reason about them without inventing numbers. The states are of three kinds: machine, gateway and fault.

Table 7. Operating states and expected telemetry

State	Kind	Expected telemetry	Source / tag
Running, outside working stroke	Machine	I about 88 A with 4 A ripple; P 4.5 bar; V breathing 1.0 to 1.7 mm/s	[P]
Running, working stroke (p 0.38 to 0.60)	Machine	I rises up to +62 A (peaks about 150 A); V swells (structural energy 62.5 to 150 Hz); P unchanged	[P]
Stopped / main motor off	Machine	I near 0 A, V near 0 mm/s, P falls (pump stops) [E]. Not modelled, so do not compare with the healthy band	[E]
Start-up and warm-up	Machine	Inrush current above running level, P builds up within seconds, cold-oil pressure above normal [E]	[E]
Idle / slide stopped, motor running	Machine	I at idle level about 88 A [P]; no working-stroke pulse (not modelled as a separate state)	[P] / [E]
Die change, setup, jog	Machine	Not modelled. Suppress diagnosis; load and vibration unrepresentative [E]	[E]
Gateway online, status online	Gateway	telemetry every 500 ms, spectrum every 2 s, synced true	[P]
Stale	Gateway	No telemetry for max(5 s, 4 x intervalMs): 5 s at the default interval	[P]
Offline	Gateway	status offline (Last Will about 15 s after power loss)	[P]
Rebooted	Gateway	New bootId, seq back at 0; simulated fault cleared by cmd/reboot	[P]
Fault ramp (NORMAL, CLOGGED_FILTER, BEARING_WEAR)	Fault	r rises linearly from 0 to 1 over rampSec then holds; NORMAL resets	[P]

3.1 State rules used by the diagnosis

- IF status is not online or the device is stale THEN do not diagnose mechanical faults; classify LS and follow the data-path rules (Document 04, Section 21).
- IF the machine is stopped, in setup or in die change [E] THEN suppress limit-based diagnosis; the repository has no state signal, so the agent must obtain it from the operator or controller.
- IF a cmd/scenario acknowledgement (ack ok true) was seen within the window THEN the abnormality is an injected simulation event; record it as such.
- IF the machine runs normally THEN evaluate all channels with the limits in Section 6.

3.2 Gateway health as a precondition

Table 8. Gateway health fields (diag, every 15 s)

Field	Healthy expectation	Abnormal meaning
uptimeS	Increases	Reset to small value: reboot
rssi	Stable (example -58 dBm)	Falling: Wi-Fi quality, gaps in seq
heapFree / heapMin	Stable minimum over a 30 minute soak	Falling minimum: gateway memory leak
reset	Reason of last reset (example POWERON)	Unexpected reset reasons
wifiReconnects, mqttReconnects, restarts, publishFailures	Low and stable	Rising: link or broker problem; data gaps
tls, insecure	tls true, insecure false in the contract example	insecure true: not a production-grade link

4. Measurement Definitions

4.1 Scalar channels

Table 9. Channel catalog (firmware/catalog.json)

#	Channel key	Label	Type	Unit	Min	Max	Decimals	warn	alarm
1	MAIN_MOTOR_CURRENT	Main motor current	current	A	0	250	1	warnHigh 175	alarmHigh 195
2	LUBE_OIL_PRESSURE	Lube oil pressure	pressure	bar	0	8	2	warnLow 3.6	alarmLow 3.0
3	BEARING_VIBRATION_RMS	Bearing vibration RMS	vibration	mm/s	0	12	2	warnHigh 2.5	alarmHigh 4.0

All limits are [P]. Values are clamped to the catalog minimum and maximum. Not defined in the repository: low limits for current and vibration, high limits for pressure, transducer part numbers, accuracy, calibration and sampling hardware [P].

4.2 Spectrum

Table 10. BEARING_SPECTRUM layout

Field	Value	Note
Bins	64 (amp.length must equal 64)	Messages with another length are dropped
startHz / stepHz	0 / 12.5	Bin i covers 12.5 x i Hz; span 0 to 787.5 Hz (runbook: 0 to 800 Hz)
Interval	2000 ms	One spectrum per four telemetry ticks
fundamentalHz	24.7	Lets a viewer mark 1X, 2X ... and lets the agent test harmonics
Unit	mm/s	Same unit as the RMS
Bin of 1X / 2X	Bin 2 (25 Hz) / bin 4 (50 Hz)	Resolution of 12.5 Hz means 24.7 Hz falls in bin 2 [D]

CONSISTENCY CHECK: BEARING_VIBRATION_RMS is the square root of the sum of the squared spectrum bins [P] (a repository test asserts agreement to 1e-9). If a received RMS and the latest spectrum disagree materially, suspect a data-path or decoding fault (LS).

4.3 Sampling facts that affect interpretation

Table 11. Sampling and timing

Fact	Value	Consequence
Telemetry rate	2 Hz (500 ms)	9.2 samples per 4.6 s stroke; the sample phase relative to the stroke repeats every 5 strokes (23 s, 46 samples) [D]
Samples inside the working stroke	About 2 per stroke (1.01 s window)	A single 500 ms sample can miss the current peak; use the stroke peak over several strokes [D]
Current ripple	4 A, eight cycles per stroke = 1.74 Hz	Above the 1 Hz Nyquist limit of a 2 Hz series: it aliases to about 0.26 Hz in the published samples [E]
Pump pulsation	1 Hz	Exactly at the Nyquist limit: the sampled amplitude depends on the sample phase, so use 10 s means [E]
Persistence	One last sample per channel per second stored	History is served bucketed; 48 h TTL [P]
Timestamps	SNTP epoch ms; if synced false or off by more than 2 minutes the backend stamps its own receive time	Do not trust ts ordering when synced is false [P]

5. Normal Operating Signature

The model is identical in the firmware (simulation.h) and in the JavaScript simulator (sim-models.mjs) [P]. p is the stroke phase and load the working-stroke pulse sin-squared($\pi(p - 0.38) / 0.22$) for $0.38 < p < 0.60$.

Table 12. Healthy model equations [P]

current	= 88.0 + 62.0*load + 4.0*sin(16*pi*p) + 1.8*gaussian()	[A]
pressure	= 4.5 + 0.12*sin(2*pi*t) + 0.03*gaussian()	[bar]
rms	= sqrt(sum of squared spectrum bins)	[mm/s]

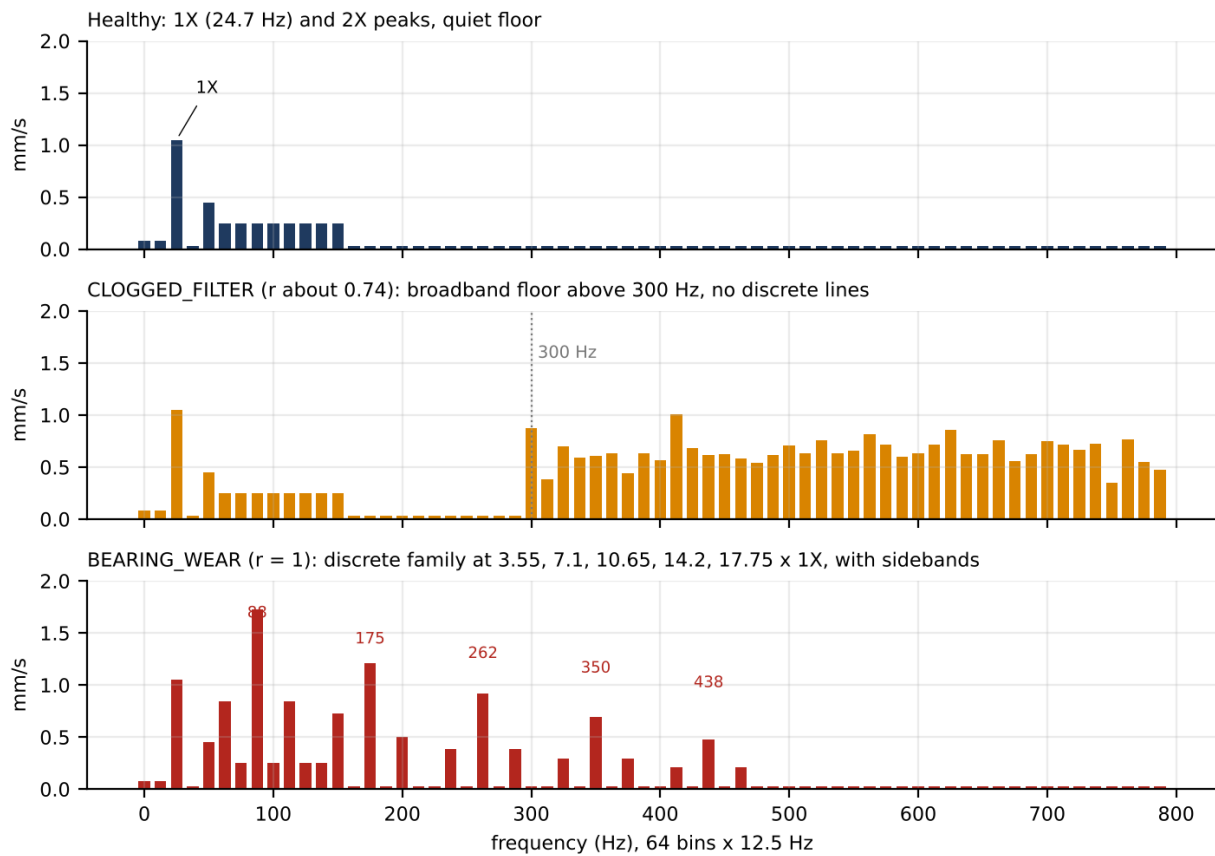
Table 13. Healthy behaviour per channel

Channel	Healthy behaviour	Observed 600 s (seed 1)	Margin to warning	Tag
MAIN_MOTOR_CURRENT	Idle about 88 A, 4 A ripple, 1.8 A noise, pulse up to +62 A in the working stroke	79.4 to 152.4 A, mean 94.8 A	Peak 152.4 A against 175 A: 22.6 A (13 percent of the limit)	[P] / [D]
LUBE_OIL_PRESSURE	4.5 bar, 0.12 bar pump pulsation at 1 Hz, 0.03 bar noise; single readings wander about 0.1 bar	4.40 to 4.59 bar, mean 4.50 bar	Minimum 4.40 bar against 3.6 bar: 0.80 bar	[P] / [D]
BEARING_VIBRATION_RMS	Breathes with the stroke between about 1.0 and 1.7 mm/s, mean near 1.3 mm/s (model constant 1.31)	1.18 to 1.72 mm/s, mean 1.31 mm/s	Peak 1.72 mm/s against 2.5 mm/s: 0.78 mm/s	[P] / [D]
Status	Always normal on all three channels over 600 s (asserted by test)	normal	Not applicable	[P]

5.1 Healthy spectrum signature

Table 14. Healthy BEARING_SPECTRUM features

Feature	Frequency (bin)	Amplitude	Meaning	Tag
1X peak	24.7 Hz (bin 2)	About 1.05 mm/s	Strongest bin; a test asserts the largest bin is bin 2	[P]
2X peak	About 49 Hz (bin 4)	About 0.45 mm/s	Second shaft harmonic	[P]
Stroke-rate band	Bins 0 and 1	About 0.08 mm/s	Energy at the stroke rate	[P]
Structural impact energy	62.5 to 150 Hz (bins 5 to 12)	0.25 x (0.4 + 1.2 x load) mm/s	Swells during the working stroke	[P]
Quiet floor	All other bins	About 0.03 mm/s	Healthy HB300 = 0.030 mm/s	[P] / [D]



Schematic rendering of the structure described in the repository; amplitudes are illustrative, not recorded data.

Figure 2. Healthy spectrum (top) and the two fault shapes. The healthy peak is at 1X; the clogged filter adds a flat floor above 300 Hz; bearing wear adds a harmonic family at non-integer orders. Schematic [I].

5.2 Example values by state

Table 15. Four consecutive 500 ms samples (same stroke phase, outside the working stroke) [D]

State	MAIN_MOTOR_CURRENT (A)	LUBE_OIL_PRESSURE (bar)	BEARING_VIBRATION_RMS (mm/s)
NORMAL	91.4, 90.4, 93.6, 95.1	4.49, 4.49, 4.51, 4.54	1.28, 1.34, 1.27, 1.27
CLOGGED_FILTER r = 0.5	88.0, 92.0, 92.9, 93.7	3.44, 3.48, 3.48, 3.50	3.19, 3.08, 3.07, 3.14
CLOGGED_FILTER r = 1	89.3, 93.3, 94.2, 95.1	2.40, 2.44, 2.48, 2.47	5.04, 4.89, 4.87, 4.97
BEARING_WEAR r = 0.5	89.3, 95.4, 93.6, 97.6	4.48, 4.49, 4.51, 4.50	2.12, 2.09, 2.10, 2.08
BEARING_WEAR r = 1	92.7, 99.0, 97.2, 101.3	4.48, 4.49, 4.51, 4.50	2.92, 2.87, 2.87, 2.86

6. Threshold System

Three of the four levels below exist in the repository; two are added here. The platform classifies each sample as normal, warn or alarm [P]. This knowledge base adds a Watch level before warning and a Critical level beyond alarm, both [I], because a plain threshold alarm is not enough (Document 02).

6.1 Level model

Table 16. Levels

Level	Name	Origin	Meaning	Typical response
L0	Normal	[P]	Inside the healthy band	None
L1	Watch	[I]	Feature has moved from baseline but no limit is reached	Trend, add to next planned inspection
L2	Warning	[P] warn limits	Warn limit reached on the persistence rule	Investigate at next planned stop; lubrication first when mixed
L3	Alarm	[P] alarm limits	Alarm limit reached on the persistence rule	Intervene this shift; plan the repair
L4	Critical	[I]	Value beyond the alarm limit by a margin, or rapid progression	Stop in a controlled way and inspect
LS	Sensor / data suspect	[E] / [I]	Signal or data path implausible	Resolve data and sensor first; do not diagnose a mechanical fault

6.2 Master limits: Minimum, Normal, Watch, Warning, Alarm, Critical

Table 17. Master threshold table (evaluated on 10 s window means unless stated)

Parameter	Minimum	Normal	Watch [I]	Warning [P]	Alarm [P]	Critical [I]	Unit	Action at warning or above
MAIN_MOTOR_CURRENT, stroke peak lpk	0 (catalog); below 60 while cycling = LS [I]	79 to 153 (sample band) [D]	above 160	at or above 175	at or above 195	at or above 220	A	Check drive and load; compare friction trend with V and P (Doc 03)
MAIN_MOTOR_CURRENT, 60 s mean l60	0	94.8 baseline, band 90 to 99 [I]	above 99.5 (1.05 x baseline)	not defined	not defined	not defined	A	Corroborating feature only: bearing wear raises it about 8 percent
LUBE_OIL_PRESSURE, P10 / P60	0 (catalog)	4.40 to 4.59 sample; P60 4.35 to 4.65 [I]	P60 at or below 4.30	at or below 3.6	at or below 3.0	at or below 2.5	bar	Replace lube filter after confirming trend and spectrum shape
BEARING_VIBRATION_RMS, V10	0 (no low limit); flat zero = LS	1.18 to 1.72 sample; V60 1.2 to 1.45 [D]/[I]	V60 at or above 1.50	at or above 2.5	at or above 4.0	at or above 7.0	mm/s	Check spectrum shape: broadband versus discrete lines
Spectrum HB300	0	0.030 healthy [D]	at or above 0.12	8 x healthy (0.24) [P] test	not defined	not defined	mm/s	Broadband floor means lubrication first
Spectrum PMR300	not defined	Healthy value not specified in repository	not defined	above 3 with discrete lines = bearing family [P] test	not defined	not defined	ratio	Discrete lines mean bearing inspection

The repository defines only the Warning and Alarm columns for the three scalar channels. Minimum is the catalog minimum. Watch and Critical are proposals [I]. Spectrum rows are feature tests taken from the repository's own test suite [P], not alarm limits.

6.3 Persistence, hysteresis and debounce

Today's platform logic classifies each sample, reaching a limit counts as breached ($\geq$ for a high limit, $\leq$ for a low limit), alarm wins over warn, and there is no hysteresis, delay or latching [P]. A cyclic signal therefore flickers across a limit. The planned detector adds hysteresis (beyond the alarm limit for 10 s opens, back inside for 10 s clears), drift detection and a debounce of one open investigation per machine with a 10 minute cooldown [P] planned. This document adopts the same 10 s timing for every level.

Table 18. Open and clear rules per level [I]

Channel	Warning opens	Warning clears	Alarm opens	Alarm clears	Critical opens
MAIN_MOTOR_CURRENT	l _{pk} at or above 175 A in 2 of 3 strokes	l _{pk} below 170 A for 10 s	l _{pk} at or above 195 A in 2 of 3 strokes	l _{pk} below 190 A for 10 s	Any 10 s mean at or above 220 A, or a sample at or above 240 A
LUBE_OIL_PRESSURE	P10 at or below 3.6 bar for 10 s	P10 above 3.8 bar for 10 s	P10 at or below 3.0 bar for 10 s	P10 above 3.2 bar for 10 s	P10 at or below 2.5 bar for 10 s
BEARING_VIBRATION_RMS	V10 at or above 2.5 mm/s for 10 s	V10 below 2.3 mm/s for 10 s	V10 at or above 4.0 mm/s for 10 s	V10 below 3.7 mm/s for 10 s	V10 at or above 7.0 mm/s for 10 s

Why a 10 s mean: the vibration RMS breathes with the 4.6 s stroke and the pressure pulsates at 1 Hz, so a 10 s window (about two strokes) averages both cycles. A per-sample rule would flicker, as the platform does for BEARING_WEAR at $r = 0.6$ to 0.8 (Section 15).

6.4 Status presentation in the twin

Table 19. How status shows [P]

Status	Mesh tint	Channel marker	Notes
normal	None (flashes green when first bound)	Circle	Default
warn	Amber #d98400, steady	Triangle	Per sample, no latching
alarm	Red #b3261e, pulsing	Square	Status message announced
agent highlight	Tone above alarm priority	Built, unused until Phase 5	Example command highlights the mesh labelled Lubrication unit

7. Feature-Based Thresholds

Limits on the raw channels alone miss a bearing that wears while RMS stays under 4.0 mm/s [P]. Features computed from windows, trend and spectrum shape catch it. The ratios below follow from the simulation equations [D]; the numeric bands are [I].

7.1 Feature definitions

Table 20. Features

Feature	Definition	Healthy value	Watch band [I]	Source
V60	Mean of BEARING_VIBRATION_RMS over 60 s	1.31 mm/s	1.50 to 2.49 mm/s	[D]
V ratio	V60 / 1.31 mm/s (the channel's own baseline)	1.00	1.15 or more	[D]
P60	Mean of LUBE_OIL_PRESSURE over 60 s	4.50 bar	4.30 down to 3.61 bar	[D]
P ratio	P60 / 4.50 bar	1.00	0.955 or less	[D]
I60 / I ratio	Mean current over 60 s / 94.8 A	94.8 A / 1.00	1.05 or more	[D]
Ipk	Maximum current in one 4.6 s stroke	About 150 A	Above 160 A	[D]
PF60	Share of samples at or above 2.5 mm/s in 60 s	0 percent	Above 0 percent	[D]
HB300	Mean amplitude of bins 24 to 63 (300 to 787.5 Hz)	0.030 mm/s	0.12 mm/s or more	[P] test uses 8 x healthy
PMR300	Peak divided by median of bins above 300 Hz	Not specified in repository	Below 2.5 flat, above 3 peaked	[P] tests
Defect lines	Amplitude at bins 7, 14, 21, 28, 35 relative to healthy	Healthy	At least 4 x healthy in each bin	[P] test

7.2 Severity index from the channels

Because the ramp r is hidden ground truth, the diagnosis estimates severity from the channels. For CLOGGED_FILTER: r from pressure = $(4.50 - P60) / 2.03$; r from vibration = $(V60 - 1.31) / 3.68$. For BEARING_WEAR: r from vibration = $(V60 - 1.31) / 1.58$; r from current = $(I60 - 94.8) / 7.6$. Agreement between the two estimates (within 0.15 [I]) supports the hypothesis [D].

Table 21. Estimated severity versus feature values [D]

Estimated r	CLOGGED: P60 (bar)	CLOGGED: V60 (mm/s)	CLOGGED: I60 (A)	BEARING: V60 (mm/s)	BEARING: I60 (A)
0.00	4.50	1.31	94.8	1.31	94.8
0.10	4.30	1.68	95.1	1.47	95.6
0.20	4.09	2.05	95.4	1.63	96.3
0.30	3.89	2.41	95.7	1.78	97.1
0.40	3.69	2.78	95.9	1.94	97.8
0.50	3.49	3.15	96.2	2.10	98.6
0.60	3.29	3.52	96.5	2.26	99.4
0.70	3.08	3.89	96.8	2.42	100.1
0.80	2.88	4.25	97.1	2.57	100.9
0.90	2.68	4.62	97.4	2.73	101.6
1.00	2.48	4.99	97.6	2.89	102.4

The numbers come from the equations (pressure and current) and from a linear fit to the 600 s derived table (vibration, within about 0.05 mm/s of the tabulated means) [D].

8. Lube Oil Pressure: Detailed Standards

8.1 Interpretation of LUBE_OIL_PRESSURE

Table 22. Pressure bands

Band (P10 or P60)	Level	Typical meaning	Evidence to check	Action
4.35 to 4.65 bar	L0	Healthy circuit, pump pulsation 0.12 bar	None	None
4.30 to 3.61 bar (P60 at or below 4.30)	L1	Early clog or a slow loss of pressure; V rises 3.68 mm/s per unit r	V60, HB300, spectrum floor	Trend; plan filter inspection
3.6 to 3.01 bar	L2	Filter clogged, r about 0.44 to 0.74 in the model	V above 2.5, broadband floor	Replace filter at next stop
3.0 to 2.51 bar	L3	Severe restriction, r about 0.74 to 0.99	V above 4.0 mm/s	Replace filter this shift
2.5 bar or lower	L4	Oil starvation of the bearings likely	V over 4.5 mm/s, current rise	Controlled stop
Rising above the healthy band [E]	LS / other	Blockage after the transducer or a stuck relief valve: not modelled; a different fault [P]	Pressure above 5.0 bar with normal vibration	Inspect nozzles and lines
Flat value with no pulsation	LS	Transducer or channel fault: healthy P has 0.03 bar noise and 0.12 bar ripple [P]	Std dev near 0	Check sensor and data path
Zero	LS	Pump stopped, sensor disconnected or decoding fault	Gateway and state	Check state first

8.2 Pressure notes

- Single readings wander by about 0.1 bar from pump pulsation, so never act on one reading; use the hourly trend and window means [P].
- Pressure is independent of load (no term in the stroke phase), so a pressure change is not explained by the working stroke [P].
- The pressure falls up to 45 percent at full severity: 4.5 to about 2.47 bar [P].
- Warning (3.6 bar) is reached at r of about 0.44 and alarm (3.0 bar) at r of about 0.74 [P].

9. Bearing Vibration: Detailed Standards

9.1 Interpretation of BEARING_VIBRATION_RMS

Table 23. Vibration bands

Band (V10 or V60)	Level	Typical meaning	Evidence to check	Action
1.0 to 1.7 mm/s (V60 1.2 to 1.45)	L0	Healthy breathing with the stroke	None	None
V60 1.50 to 2.49 mm/s	L1	Rising baseline; CLOGGED_FILTER r above 0.05 or BEARING_WEAR r above 0.12	P60, spectrum shape, I60	Trend; check pressure first
2.5 to 3.99 mm/s	L2	Both faults possible: clog at r about 0.33 to 0.74, bearing at r about 0.8 or more	Broadband versus discrete lines	Lubrication first; inspect at next stop
4.0 to 6.99 mm/s	L3	Alarm; only the clogged filter reaches it (about 5.0 mm/s at r = 1)	P60 below 3.0 bar	Filter swap now
7.0 mm/s or more	L4	Severe: no simulated fault reaches it; treat as real damage or sensor fault	Mount, cable, impact events	Controlled stop
Flat zero or saturated at 12 mm/s	LS	Sensor or decoding fault	Spectrum agreement with RMS	Check sensor

BEARING_WEAR peaks at about 3.2 mm/s at full severity and never reaches the 4.0 mm/s alarm [P] (asserted by test). A worn bearing can therefore be badly damaged at level L2 or below. Use trend and shape, not the alarm.

10. Motor Current: Detailed Standards

10.1 Interpretation of MAIN_MOTOR_CURRENT

Table 24. Current bands

Band	Level	Typical meaning	Evidence to check	Action
Ipk up to 153 A, I60 94.8 A	L0	Healthy working stroke	None	None
Ipk 160 to 174 A or I60 above 99.5 A	L1	Friction or load rise; bearing wear at r above 0.62 gives I60 of about 99.5 A	V60 and spectrum lines	Trend
Ipk 175 to 194 A	L2	Overload or strong friction; neither simulated fault reaches it (max 164.1 A)	Die, process, drive	Check load and drive
Ipk 195 to 219 A	L3	Heavy overload	Stroke profile, die setup	Reduce load; inspect
Ipk 220 A or more	L4	Possible jam or short	Drive protection	Stop
I near 0 A with press cycling, or flat	LS	Current sensor or data fault; motor off	State	Check state and sensor

Current is a weak, late and slow corroborating channel in this model: CLOGGED_FILTER raises it about 3 percent (to about 97.6 A mean) and BEARING_WEAR about 8 percent (to 102.3 A mean, maximum 164.1 A), and neither reaches the 175 A warning [P]. Its value is in the ratio to vibration (Section 16.3).

11. Spectrum Standards

Table 25. Spectrum bands and their use

Band	Bins	Frequency	Healthy content	Fault content
Stroke rate	0 to 1	0 to 12.5 Hz	About 0.08 mm/s	Not specifically modified
Shaft 1X	2	About 25 Hz	1.05 mm/s peak	Sidebands of bearing lines at plus and minus 1X (24.7 Hz)
Shaft 2X	4	About 50 Hz	0.45 mm/s	None
Structural band	5 to 12	62.5 to 150 Hz	0.25 x (0.4 + 1.2 x load) mm/s	Bearing lines at 88 and 175 Hz sit near or above it
Mid band	13 to 23	162.5 to 287.5 Hz	About 0.03 mm/s	Bearing lines at 175, 263 Hz
High band (HB300)	24 to 63	300 to 787.5 Hz	0.030 mm/s floor	Clog: flat floor 0.28 to 0.75 mm/s; bearing: lines at 351 and 439 Hz

Table 26. Bearing defect orders and bins [P] / [D]

Order x 1X	Frequency (Hz)	Bin	Relative weight	Sidebands
3.55	87.7	7 (87.5 Hz)	1.00	Plus and minus 1X at 0.4 of the carrier (about 2 bins, 25 Hz)
7.1	175.4	14 (175 Hz)	0.80	Same
10.65	263.1	21 (262.5 Hz)	0.60	Same
14.2	350.7	28 (350 Hz)	0.45	Same
17.75	438.4	35 (437.5 Hz)	0.30	Same

The orders are non-integer multiples of the shaft frequency, which is the signature of a rolling-element outer-race defect [E]; the repository does not state the bearing type or designation [P]. A 12.5 Hz bin spans less than the 24.7 Hz sideband spacing, so the sidebands appear two bins either side [D].

Table 27. Spectrum feature tests from the repository [P]

Test	Healthy	CLOGGED_FILTER	BEARING_WEAR
High-band mean above 300 Hz	0.030 mm/s	More than 8 x healthy (above 0.24 mm/s); 0.283 at r = 0.25, 0.749 at r = 1	Not a discriminator
Peak-to-median above 300 Hz	Not specified in repository	Below 2.5 (flat, no discrete family)	Above 3 (discrete peaks)
Defect bins 7, 14, 21, 28, 35	Healthy amplitude	Not raised as a family	Each more than 4 x healthy
Largest bin	Bin 2 (1X)	Not required	Not required

12. Startup and Shutdown Behaviour

The simulator has no start-up or shutdown state: after a boot or a cmd/reboot the model runs with $r = 0$ (NORMAL) at once, and the gateway publishes status online [P]. A real press has start-up and shutdown behaviour that the repository does not specify. The table is engineering reference [E] and must be replaced by plant data.

Table 28. Start-up and shutdown behaviour

Phase	MAIN_MOTOR_CURR ENT	LUBE_OIL_PRESSURE	BEARING_VIBRATION _RMS	Diagnostic treatment
Gateway boot (simulator)	Model starts at normal	Model starts at normal	Model starts at normal	Wait for status online and synced true [P]
Motor start [E]	Inrush above running level for seconds	Rises as the pump fills the lines	Run-up transient through resonance	Suppress limit checks for the start-up window set by the plant
Warm-up [E]	Slightly higher friction with cold oil	Higher with cold, viscous oil	Slightly different	Compare with a warm baseline only after warm-up
Steady running	See Section 5	See Section 5	See Section 5	Full diagnosis
Controlled shutdown [E]	Falls to zero	Falls as the pump stops; no alarm expected	Falls to near zero	Suppress low-pressure alarms when the motor is off
Power loss	status offline about 15 s later [P]	Last values retained until stale	Same	Classify as data state, not a fault

IF the main motor is off THEN pressure below 3.0 bar is not a lubrication alarm. The repository has no motor-run signal, so the agent must obtain the machine state from outside the three channels (current near zero is a hint, not proof) [E].

13. Load-Dependent Behaviour

Table 29. Dependence of each channel on load (stroke phase)

Channel	Depends on load?	Behaviour	Evaluation rule
MAIN_MOTOR_CURR ENT	Yes, strongly	+62 A sin-squared pulse over p 0.38 to 0.60; ripple 4 A across the stroke	Evaluate stroke peak I_{pk} and the 60 s mean separately; compare the idle level to 88 A
BEARING_VIBRATION _RMS	Yes, moderately	Breathes 1.0 to 1.7 mm/s; structural energy 62.5 to 150 Hz swells with load	Use V60 and compare against the healthy breathing band; do not judge one stroke peak
LUBE_OIL_PRESSURE	No	4.5 bar plus 1 Hz pulsation	Evaluate directly on P10 and P60
BEARING_SPECTRUM	Partly	Impact band varies with load; 1X and 2X stable	Use the 2 s spectrum; bins above 300 Hz are load independent in the healthy model

Table 30. Stroke phase table for the 4.6 s cycle [P] / [D]

Phase p	Time in stroke (s)	Load	Expected current (noise-free)	Note
0.00 to 0.38	0 to 1.75	0	88 A plus 4 A ripple	Non-working part
0.38 to 0.49	1.75 to 2.25	Rising	Up to 150 A	Pulse rises
0.49	2.25	1.0	About 150 A (88 + 62)	Peak load
0.49 to 0.60	2.25 to 2.76	Falling	Down to 88 A	Pulse falls
0.60 to 1.00	2.76 to 4.6	0	88 A plus ripple	Return

IF the current peak per stroke exceeds 160 A while V60 and P60 are healthy THEN suspect a process or die load change (Document 03 rule for load) rather than a lubrication fault [E].

14. Environmental Influences

The repository does not model ambient or oil temperature, supply voltage, die or material changes, or foundation vibration, and it gives no environmental limits [P]. The influences below are engineering reference [E]; they explain why a baseline must be recomputed and why a reading can move without a fault.

Table 31. Influences and their expected effect

Influence	Channel affected	Expected effect	Treatment
Oil temperature or viscosity	LUBE_OIL_PRESS URE	Colder oil raises pressure; hot thin oil lowers it	Compare to the same operating temperature; temperature is not measured
Ambient temperature	P, I	Moves oil temperature and friction slightly	Seasonal baseline update
Supply voltage variation	I	Current changes for the same load	Check supply if I moves without V or P
Die weight, material or blank thickness	I	Changes the working-stroke pulse and the stroke peak	Re-baseline after a die change
Neighbouring machine or foundation vibration	V	Adds broadband or line energy	Compare with the stopped-machine reading
Electrical noise near the vibration cable	V, spectrum	Adds spikes or mains-frequency lines	Check shielding
Oil contamination	P	Faster filter clogging; repository: minutes to days depending on oil cleanliness [P]	Oil analysis
Network quality	All	Gaps in seq, stale device	Data-path rules

15. Normal Transients and False Alarms

Table 32. Normal transients versus abnormal behaviour

Observation	Normal transient?	Why	Rule
Single current sample near 150 A	Yes	Working-stroke pulse peaks at $88 + 62 = 150$ A [P]	Judge Ipk against 175 A
Pressure reading 0.1 bar from 4.5	Yes	Pump pulsation 0.12 bar at 1 Hz [P]	Use window means
Vibration RMS between 1.0 and 1.7 mm/s	Yes	Breathes with the stroke [P]	Use V60
Vibration single sample at 2.5 mm/s while V60 is 1.3	Unlikely in the healthy model (max 1.72 mm/s)	Healthy 600 s maximum 1.72 mm/s [D]	Treat as spike or data error; confirm with the next spectrum
V flickers across 2.5 mm/s	Abnormal	BEARING_WEAR at r about 0.6 to 0.8 flickers across warn [P]	Use PF60 and V60
Pressure drops under 3.6 bar for one sample	No, unless P10 confirms	One sample at 0.03 bar noise cannot cross 3.6 from 4.5	Check P10
Values change after cmd/scenario ack	Injected event	A fault command is a simulation action [P]	Mark as injected
All three channels flat	No	Healthy model has noise on every channel	LS

16. Rate of Change and Time to Threshold

16.1 Rates per unit ramp

Table 33. Rates [D]

Fault	Channel	Change per unit r	At rampSec = 120 s (per minute)	At rampSec = 600 s (per minute)
CLOGGED_FILTER	P60	-2.03 bar	-1.01 bar/min	-0.20 bar/min
CLOGGED_FILTER	V60	+3.68 mm/s	+1.84 mm/s/min	+0.37 mm/s/min
CLOGGED_FILTER	I60	+2.8 A	+1.4 A/min	+0.28 A/min
BEARING_WEAR	V60	+1.58 mm/s	+0.79 mm/s/min	+0.16 mm/s/min
BEARING_WEAR	I60	+7.6 A	+3.8 A/min	+0.76 A/min
BEARING_WEAR	P60	0	0	0

16.2 Time to threshold by ramp duration

Table 34. Time to threshold after the start of the ramp [D]

Fault	Event	r	60 s ramp	120 s ramp (default)	300 s ramp
CLOGGED_FILTER	V60 at 1.50 mm/s (Watch)	0.05	3 s	6 s	15 s
CLOGGED_FILTER	First samples at or over 2.5 mm/s	0.25	15 s	30 s	75 s
CLOGGED_FILTER	V60 at 2.5 mm/s	0.33	20 s	40 s	99 s
CLOGGED_FILTER	P60 at 3.6 bar (warn)	0.44	26 s	53 s	132 s
CLOGGED_FILTER	P60 at 3.0 bar and V60 at 4.0 mm/s (alarm)	0.74	44 s	89 s	222 s
BEARING_WEAR	V60 at 1.50 mm/s (Watch)	0.12	7 s	14 s	36 s
BEARING_WEAR	First samples at or over 2.5 mm/s	0.60	36 s	72 s	180 s
BEARING_WEAR	Mostly over 2.5 mm/s (88 percent)	0.80	48 s	96 s	240 s
BEARING_WEAR	4.0 mm/s alarm	-	never reached	never reached	never reached

The demo script gives a one-minute-ramp timeline for the clogged filter: about 20 s vibration peaks cross warn, about 27 s pressure drops under 3.6 bar, about 45 s pressure under 3.0 bar and vibration over 4.0 mm/s [P]; the table above is consistent with it. In a real plant a clogging filter develops over minutes to days depending on oil cleanliness [P], so scale by the observed slope, not by seconds.

16.3 Cross-channel rate ratios

Table 35. Ratios that separate the two faults [D]

Ratio	CLOGGED_FILTER	BEARING_WEAR	Use
Delta V60 / Delta P60	-1.8 mm/s per bar (V rises 3.68 per 2.03 bar fall)	Undefined: P60 unchanged	P falling with V rising means filter
Delta I60 / Delta V60	0.77 A per mm/s (2.8 A per 3.68 mm/s)	4.8 A per mm/s (7.6 A per 1.58 mm/s)	About six times higher for bearing wear
Delta HB300	0.03 to 0.75 mm/s, flat	Small, lines only	Broadband versus discrete
Delta V60 at full severity	+3.68 mm/s, 4.98 mean (above alarm)	+1.58 mm/s, 2.89 mean (below alarm)	Alarm reached or not

17. Abnormal Conditions and Interpretation

Table 36. Abnormal conditions and first interpretation

Condition	Threshold evidence	Interpretation	Level	Next step
P10 at or below 3.6 bar and V60 at or above 2.5 mm/s	Both	CLOGGED_FILTER likely (Document 02 F01)	L2	Confirm broadband spectrum; replace filter
P10 at or below 3.0 bar and V60 at or above 4.0 mm/s	Both alarm	CLOGGED_FILTER severe	L3	Replace filter now; inspect bearings after
V60 rising from 1.3 toward 3 mm/s with P normal	No limit reached	BEARING_WEAR drift [P]	L1 to L2	Handheld analyser; plan replacement
V60 at or above 2.5 mm/s, P normal, discrete lines	Warn	BEARING_WEAR	L2	Plan bearing inspection at next stop
Ipk at or above 175 A with V60 and P60 normal	Warn on current	Load or drive, not a modelled fault	L2	Check die, process, drive
V10 at or above 7.0 mm/s	Critical	Severe fault or sensor fault	L4	Stop; inspect
P rising above healthy	Not defined	Blockage after the transducer [P] (not modelled)	LS / other	Inspect nozzles and lines
Implausible values or gaps	None	Sensor or data fault	LS	Resolve data path

17.1 Reasoning rules

- IF pressure is normal and the spectrum shows discrete peaks THEN it is not a filter problem; use the bearing procedure [P].
- IF the evidence is mixed THEN check the lubrication first: a clogged filter is the cheap fault and it causes bearing damage if left [P].
- IF V60 climbs steadily from 1.3 toward 3 mm/s THEN treat it as drift whatever the limit says [P].
- IF P60 falls and HB300 rises together THEN lubrication starvation is the leading hypothesis [P] [D].

18. Alert Escalation

Table 37. Escalation matrix [I]

Level	Condition	Notify	Time to act (prototype)	Action authority
L0	Healthy	None	None	None
L1	Watch feature, no limit	Maintenance planner (information)	Next planned inspection	Planner
L2	Warn limit on persistence rule	Maintenance technician	Next planned stop	Technician plans work
L3	Alarm limit on persistence rule	Shift supervisor and maintenance	This shift	Supervisor decides on stop
L4	Critical or rapid progression	Production and maintenance manager	Immediate	Controlled stop
LS	Sensor or data suspect	Instrumentation / IT	Within shift	Technician; no mechanical work until resolved

Table 38. Escalation modifiers

Modifier	Effect
Two or more channels at L2 or above	Raise one level (not above L4)
Time to alarm shorter than 15 minutes of real time by the 60 s slope [I]	Raise one level
Mixed evidence between the two faults	Check lubrication first, keep level
Recent maintenance on the circuit	Add post-maintenance verification (Document 03)
Debounce	One open investigation per machine; 10 minute cooldown [P] planned
Agent authority	The agent cannot act on the machine; it produces a report and user-interface commands only [P] planned

19. Data Quality and Baseline Establishment

19.1 Baseline procedure

Table 39. Baseline establishment [I]

Step	Action	Acceptance
1	Confirm status online, synced true, no seq gaps over the window	No gaps
2	Collect at least 10 minutes in steady running (about 130 strokes)	Window complete
3	Compute I60, Ipk, P60, V60, HB300 and PMR300	Values in healthy bands
4	Store spectrum bins 0 to 63 (mean over the window)	1X is the largest bin
5	Record the baseline with date, state and operator	Signed baseline
6	Recompute after any repair, die change or oil change	Updated baseline

19.2 Data-quality checks

Table 40. Checks before interpretation

Check	Pass	Fail means
status	online	Offline: data state
Staleness	Telemetry within max(5 s, 4 x intervalMs)	Stale
seq continuity	No gaps	Packet loss; use fewer samples
bootId and seq	No unexpected reset	Reboot; fault cleared in simulation
Variation	Std dev above zero on every channel	Flat signal: LS
Range	Inside catalog min and max	Clamped: possible saturation
RMS versus spectrum	Agree	Decoding or data fault
Healthy-model plausibility	I at least 79 A at idle	Implausible

20. Summary, Safety, References and Revision

20.1 Key thresholds

Table 41. Key thresholds at a glance

Channel	Normal	Warning [P]	Alarm [P]	Critical [I]	Unit
MAIN_MOTOR_CURRENT	79 to 153 sample; mean 94.8	at or above 175	at or above 195	at or above 220	A
LUBE_OIL_PRESSURE	4.40 to 4.59 sample; mean 4.50	at or below 3.6	at or below 3.0	at or below 2.5	bar
BEARING_VIBRATION_RMS	1.18 to 1.72 sample; mean 1.31	at or above 2.5	at or above 4.0	at or above 7.0	mm/s

20.2 Safety and escalation notes

This document is a capstone prototype and does not replace the OEM manual, the plant lock-out and tag-out procedure or site safety rules. Never act on a limit in this document without confirming it against the real equipment documents. The diagnosis agent cannot act on the machine.

20.3 References and engineering basis

- Repository context file CONTEXT_PRESS_STAMP_01.md and the files it cites: firmware/catalog.json, simulation.h, sim-models.mjs, docs/mqtt-contract.md, the two press runbooks, channelStatus.js, simulator.test.js, demo scripts and the Phase 5 blueprint [P].
- General vibration severity practice for rotating machines (for example the ISO 20816 family) and general lubrication-circuit practice, used only as engineering reference for the Watch and Critical proposals [E]. No ISO zone value is claimed as a project limit.
- Illustrative values [I] are explained in Section 6 and must be replaced by plant data.

20.4 Revision notes

Table 42. Revision history

Rev	Date	Change
1.0	2026-10-08	First issue for the capstone prototype knowledge base